



Effect of Force-Rate on Continuous Kinesthetic Force Discrimination

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INTRODUCTION

Conventional haptic force discrimination tasks involve perceptual differentiation between stimulus pairs separated by a brief inter-stimulus interval (ISI).

Objective:

It is unclear whether the Force JND measured with ISI remains consistent in **Continuous Force Discrimination**.

Weber Fraction:

$$\delta = \frac{|F_t - F_r|}{|F_r|},$$

where, F_r = Reference Force, F_t = Test Force

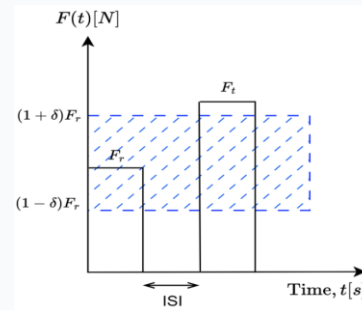


Fig. 1: Force Discrimination with ISI and corresponding deadzone

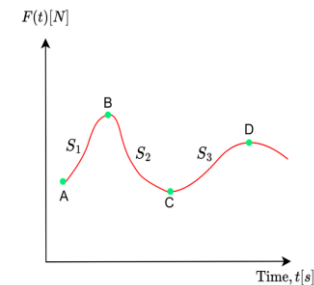


Fig. 2: Continuous Force Stimuli $F(t)$ with varying force-rates

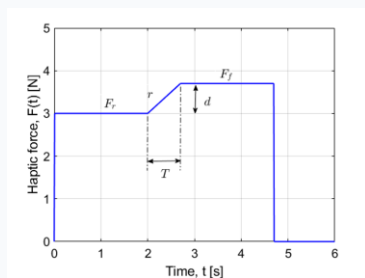


Fig. 3: Force Stimuli Transition with force-rate (r)

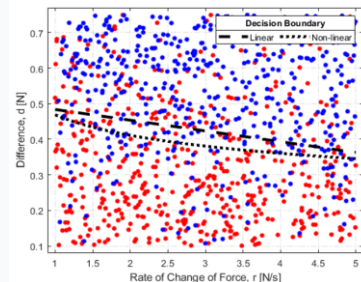


Fig. 4: Scatter Plot for one participant

METHODOLOGY

Machine learning classification-based framework

Setup & Stimuli:

PHANToM Premium 1.5 HF (1D haptic force stimulus +Z-axis).
R [1,5] N/s, D [0.1, .75] N

Data Collection: 10 users, 1000 trials per user

Paradigm: Participant is exposed to a linearly increasing 1D kinesthetic force stimulus and reacts to the change in force.

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RESULTS AND ANALYSIS

Classifier & Prediction Error

- Mean prediction error: $E^I = 27.94\%$, $E^P = 27.11\%$
- No significant difference in error ($t = 1.388$, $p = 0.198$)

Two-Way ANOVA: Factors: force-rate and perceptual boundaries

- **Significant effect of**
 - force-rate on Weber fraction
 - perceptual boundaries on Weber fraction
- **Interaction effect is not significant**

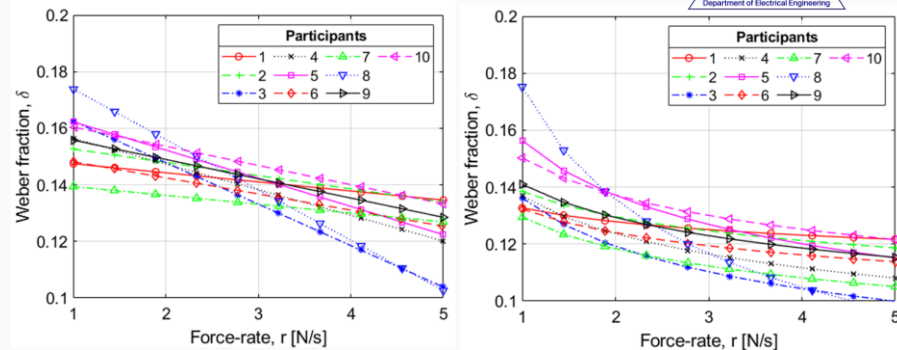


Fig. 5: (a) Linear, (b) Power function-based boundaries

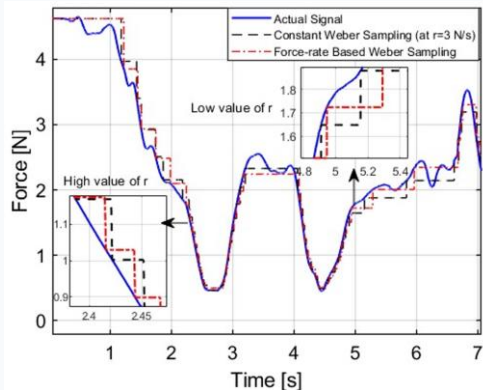


Fig. 6: Force signal & reconstructed signals using both Constant Weber fraction (at $r = 3$ N/s) and force-rate-based Weber fraction sampling

DISCUSSION & CONCLUSION

- Positive force-rate $[1-5]$ N/s $\uparrow$, Weber fraction (δ) $\downarrow$
- Indicating improved sensitivity to force changes
- Application: Perceptually adaptive haptic codecs and communication frameworks